ROCK · PAPER · SCISSORS

GAME DESIGN DOCUMENT

Version 1.0 | Desktop prototype | SDL2 + C++17

Document purpose

This document describes the game as implemented in the supplied C++ source code. Where a design intention is inferred from the implementation, it is identified as such. Proposed future features are kept separate from the current feature set.

1. Game Overview

High concept: A simple, single-player desktop version of Rock Paper Scissors. The player chooses one of three hand gestures, the computer randomly selects its own gesture, and the game resolves the round using the classic rules. Animated hand illustrations, a score display, and colour-coded results provide feedback.

Genre: Casual / party game / reflex-free strategy game.

Platform: Desktop (the source includes Linux/macOS and Windows build guidance).

Player count: One human player versus a computer opponent.

Session structure: Short, repeatable rounds with a running score. There is no defined match length or final victory condition in the current implementation.

2. Core Gameplay

2.1 Objective

Choose Rock, Paper, or Scissors to beat the computer's randomly generated choice. Wins increase the player's score; losses increase the computer's score; ties do not change either score.

2.2 Rules

- **Rock beats Scissors.**
- **Scissors beats Paper.**
- **Paper beats Rock.**
- Matching choices produce a tie.
- Every completed round compares one player choice with one computer choice.

2.3 Round flow

- The game waits for the player to choose a hand.
- On a valid choice, the player's selection is stored and the computer chooses randomly from the three options.
- The game enters an animation phase. The player's hand bounces while the computer's choice remains concealed behind a question mark.
- After the animation, the computer's hand is revealed and the round result is calculated.
- The score is updated for a win or loss. A tie leaves the score unchanged.
- The result and both choices are displayed. The player can select another hand to begin the next round.

3. Controls

Input	Action
Left mouse button	Select Rock, Paper, or Scissors; starts a round.
Mouse movement	Highlights a choice button when the pointer is over it.
Right mouse button	Resets both scores and returns the game to its idle state (when input is accepted).
Escape	Closes the game.
Window close button	Closes the game.

Note: The reset is triggered by a right-click anywhere in the game window, not by a dedicated on-screen reset button. The source only processes gameplay mouse input while the game is idle or showing a result.

4. User Interface and Visual Direction

The interface uses a fixed 800 × 600 window and a dark, high-contrast purple theme. Its layout is organised around a centred title, a score panel, two side-by-side play arenas, and three large choice buttons along the bottom.

4.1 Visual elements

- **Background:** Deep navy (#0F0C1E).
- **Panels:** Dark purple (#1C1837) with violet borders.
- **Accent:** Vivid violet (#8250FF).
- **Text:** Near-white primary text and muted purple secondary text.
- **Result colours:** Green for a player win, red for a loss, and amber for a tie.
- **Hand illustrations:** Rock, Paper, and Scissors are drawn from SDL rendering primitives rather than external image assets.
- **Buttons:** Each choice button displays a small hand illustration and a text label, with idle, hover, and pressed visual states.

4.2 Screen layout

- Top: game title.
- Below title: score panel showing YOU and CPU scores.
- Middle: player and CPU arenas, each labelled.
- Bottom: three large selection buttons arranged horizontally.
- Contextual text: idle prompt, throwing animation text, result message, choice comparison, and reset hint.

5. Animation and Feedback

- The main loop targets 60 frames per second and uses elapsed frame time for animation updates.
- The round animation lasts approximately 1.8 seconds.
- During the animation, the player's hand moves with a decaying sine-wave bounce.
- The computer's choice is hidden during the animation and then revealed.
- The reveal transition is configured for approximately 0.6 seconds.
- A changing sequence of dots is appended to the word “Throwing” during the animation.
- The final result is shown using a large, colour-coded message.

6. Game Systems

6.1 Computer opponent

The computer selects Rock, Paper, or Scissors using the C standard library random number generator. The generator is seeded from the current system time when the application starts. The current opponent does not adapt to player behaviour or use a difficulty setting.

6.2 Scoring

The game maintains two integer counters: player score and computer score. A player win increments the player counter; a player loss increments the computer counter; a tie changes neither. Scores remain available between rounds until reset or application exit. The source does not save scores to disk.

6.3 Game states

State	Purpose
IDLE	Waits for the player to make a choice and displays the starting prompt.
ANIMATING	Plays the hand-bounce animation and keeps the computer choice hidden.
RESULT	Displays the revealed choices and round outcome; accepts the next round.

7. Audio, Narrative, and Accessibility

The supplied implementation contains no audio, music, voice acting, story, characters, or narrative progression. It also does not include dedicated accessibility settings. The game communicates state primarily through text, colour, hand illustrations, and mouse interaction.

8. Technical Design

- **Language:** C++17.
- **Libraries:** SDL2 for windowing, input, and rendering; SDL2_ttf for font rendering.
- **Window:** 800 × 600 pixels.
- **Rendering:** SDL renderer with primitive shapes, lines, points, and rendered text textures.
- **Fonts:** Attempts DejaVu Sans, then Arial, then Liberation Sans paths. Font availability is platform-dependent.
- **Frame target:** 60 FPS (with renderer present-vsync also enabled).
- **External game assets:** No separate image or audio assets are required by the supplied source.

8.1 Build requirements

The source comments specify SDL2, SDL2_ttf, and a C++17-capable compiler. The provided build examples are:

Linux/macOS: `g++ rps_game.cpp -o rps_game $(sdl2-config --cflags --libs) -lSDL2_ttf -std=c++17`

Windows: Use the supplied vcpkg dependencies (sdl2 and sdl2-ttf) and link SDL2.lib, SDL2_ttf.lib, and SDL2main.lib as shown in the source header.

9. Current Scope and Limitations

- Single-player only; no local or online multiplayer.
- One computer opponent with random choices; no difficulty levels.
- No match limit, tournament mode, or overall match winner screen.
- No persistent save data, statistics, leaderboard, or history.
- No settings menu, pause menu, or dedicated on-screen reset control.
- No audio or external visual assets.
- The window dimensions and layout are fixed in the source.

10. Suggested Future Development

The following are optional ideas, not features currently present in the supplied code:

- **Best-of match modes:** Add first-to-three, first-to-five, or custom round limits with a match-end screen.
- **Difficulty options:** Offer random, pattern-based, or adaptive computer behaviour, clearly explaining how each mode works.
- **Statistics:** Track total rounds, wins, losses, ties, and win percentage during a session.
- **Improved controls:** Add keyboard shortcuts and a visible reset button, with clear focus and hover states.
- **Settings:** Add sound, animation, and display options if those features are introduced.
- **Persistence:** Optionally save settings and high-level statistics between sessions.
- **Polish:** Add sound effects, transitions, and a more explicit match-start or match-end flow.

11. Acceptance Criteria

- The application launches into an idle screen with the title, score display, arenas, and three choice buttons.
- The player can select Rock, Paper, or Scissors using the left mouse button.
- A round begins with the player's choice shown and the computer choice concealed during the animation.
- The computer chooses one of the three valid options.
- The correct win, loss, or tie outcome is displayed for every possible pair of choices.
- The player score increments only on a win; the computer score increments only on a loss; ties do not change either score.
- The player can start another round after the result appears.
- Right-click resets the scores, and Escape or the window close control exits the application.

12. Source-Based Design Notes

This GDD is based on the supplied SDL2 C++ source. It documents implemented behaviour rather than assuming features that are not present. Product positioning, audience, and future-development ideas are design interpretations or proposals, not claims made by the code.

Primary source: User-provided Rock Paper Scissors SDL2 C++ game source code.